

VENKATA SATYA SAI PRUDHVI GUDLA

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Education

Year	Degree/Certificate	Institute	CPI/%
2025	B.Tech, Metallurgical and Materials Engineering & Minor in Computer Science Engineering	Indian Institute of Technology, Kharagpur	8.66/10
2021	APBIE(XII)	Sri Venkateswara Classes, Visakhapatnam	96.1%
2019	CBSE(X)	Delhi Public School, Visakhapatnam	95.8%

Publications

- A Amalanshu, V Nagaswamy, **G Prudhvi**, Y Sirvi, Entity Augmentation for Efficient Classification of Vertically Partitioned Data with Limited Overlap - GLOW Workshop, International Joint Conference on Artificial Intelligence, 2024
- P Gudla**, K Singh, CBL: Compact Encoding of JSON-LD Data using CBOR and Bitmaps for Web of Things-arXiv preprint

Experience

Laboratoire Hubert Curien, Jean Monnet University

Apr 2024 – Jul 2024

Charpak Lab Scholar - Research Internship | Guide: Prof. Kamal Singh

St. Etienne, France

- Worked on the Constrained Semantic Web of Things project sponsored by the French National Research Agency(ANR)
- Crafted a novel compression algorithm for JSON-LD data that provides network overhead savings up to 95.1%
- Built a Python library for efficient encoding and decoding of JSON-LD data using our novel compression algorithm

Smart Wireless Applications and Networking Lab, IIT Kharagpur

Jan 2024 – Nov 2024

Project Lead, SteelEye - Ladle Tracking System

Kharagpur, West Bengal, India

- Built a Ladle Tracking System that automates ladle number detection in steel plants using a fine-tuned YOLOv8 model and a FastAPI web application integrated with a MySQL database to deliver real-time key performance indicators
- Developed a RAG chatbot that generates SQL queries to retrieve data and answer user queries within the web app

Autonomous Ground Vehicle Research Group, IIT Kharagpur

Aug 2022 – Apr 2025

Mechatronics and Vision Team lead | Guide: Prof. Debashish Chakravarty

Kharagpur, West Bengal, India

- Engineered a novel entity augmentation technique for vertical federated learning, achieving a 69.5% accuracy on resnet18 trained on the CIFAR10 dataset with 5% entity alignment, significantly outperforming traditional VFL approaches
 - Researched extensively on 3D reconstruction methods, including Neural Radiance Fields and 3D Gaussian Splatting.
- Authored a reproducibility report: “Exploring Explicit Representations in 4D: A Comparative Analysis with HexPlane”

Projects

Smart Vision System for Quality Control

Oct 2024 – Nov 2024

- Developed a Quality Tracking System by fine-tuning the Florence2 VLM on a custom dataset, enabling OCR-based recognition of product details for consumer goods, and freshness index assessment of fruits and vegetables

Machine Learning for Modelling Interatomic Potentials

Jan 2025 – Apr 2025

- Developed a Moment Tensor Potential for scalable molecular dynamics simulations of Tungsten and Gallium Arsenide

Visual Web Agent

Jan 2025

- Engineered a multi-stage visual AI system using LangGraph and Gemini Pro Vision to autonomously navigate web UIs

Achievements

- Awarded the **Charpak Lab Scholarship 2024**, granted to 35 students nationwide from the French Institute in India
- Received the **AI4ICPS Chanakya Fellowship** for building a end-to-end distributed ladle tracking system for steel plants
- Achieved **National Finalist** position at **Smart India Hackathon 2023** as team lead for developing an end-to-end pipeline for automating real-time hot metal ladle tracking at Visakhapatnam Steel Plant through ladle number detection
- Built the gold-winning prototype using a conveyor belt system, movable Intel RealsenseD455 Depth camera, and computer vision methods such as U2NetP for volumetric weight estimation at **Inter IIT Tech Meet 11.0, Kanpur, UP**

Skills

Languages and Libraries: C, Matlab, Python, SQL, Numpy, Pandas, Matplotlib, Pyspark, Scikit-learn, TensorFlow, Pytorch, CUDA, Flower, FastAPI, Streamlit, MLflow, Langchain, Langgraph, OpenCV, Open3D, mne, NetworkX

Technologies/Frameworks: ROS, Git, Github, Docker, Kubernetes, Airflow, Kafka, Linux, AWS, Arduino, ESP32

Positions of Responsibility

Student Satellite Program IIT KGP

Nov 2023 – Apr 2025

Founding Member

Kharagpur, West Bengal

- * Executed a pivotal role in the **CubeSat** design and overseeing sensor testing, successfully implementing a LoRa RFM95W-based communication system with a spring antenna on the satellite and a Yagi antenna at the ground station

Relevant Coursework

- Advanced Digital Image Processing & Computer Vision
- Machine Learning Foundation and Applications
- Computer Architecture and Operating Systems
- Probability & Statistics
- Programming and Data Structures
- Switching Circuits and Logic Design